Nikhil Krishnaswamy

https://www.nikhilkrishnaswamy.com
https://www.signallab.ai

Areas of Specialization

Artificial Intelligence; Computational Linguistics/Natural Language Processing; Machine Learning; Human-Computer Interaction; Simulation; Computer Graphics; Spatial Cognition and Reasoning.

Appointments Held

2020-present *Assistant Professor of Computer Science*, Colorado State University, Fort Collins, CO, USA

2017-2020 *Postdoctoral Associate*, Brandeis University, Waltham, MA, USA

Education

2017 PhD in Computer Science, Brandeis University, Waltham, MA, USA
Advisor: *Prof. James Pustejovsky*. Committee: *Prof. Kenneth D. Forbus*, *Prof. Timothy J. Hickey*, *Dr. Marc Verhagen*.

2013 MA in Computational Linguistics, Brandeis University, Waltham, MA, USA

2010 BS in Computer Games Development, DePaul University, Chicago, IL, USA

Doctoral Thesis

2017 Krishnaswamy, N. (2017). *Monte-Carlo Simulation Generation Through Operationalization of Spatial Primitives*. PhD thesis, Brandeis University.
• Created VoxSim semantic event simulator and VoxML visual modeling language (now under development as an ISO standard).

1 Research

Research Funding

2023 Army Research Office Short Term Innovative Research: Embodied Computational Metacognition. Award #W911NF-23-1-0031. Funded $60,000.

2020-2026 National Science Foundation Division of Research on Learning: AI Institute: Institute for Student-AI Teaming (Term as co-PI: 2022-2026). Award #DRL 1559731. Funded $632,644.

2020-2022 National Science Foundation Division of Information & Intelligent Systems: NSF2026: EAGER: A Playground and Proposal for Growing an AGI (consultant). Award #CNS

1033932. Funded $12,000.

2018-2022 Defense Advanced Research Projects Administration Information Innovation Office: Representations of Vectors and Abstract Meanings for Information Synthesis (Active Interpretation of Disparate Alternatives Program) (Term as co-PI: 2022). Award #FA8750-18-2-0016. Funded $436,178.

Honors & Awards

2022 Best paper, International Conference on Human-Computer Interaction (HCII). *Multimodal Semantics for Affordances and Actions*.

Best interactive event, International Conference on Artificial Intelligence in Education (AIEd). *iSAT speech-based AI display for small group collaboration in classrooms*.

2020 Best demo, International Conference on Artificial Reality and Telexistence & Eurographics Symposium on Virtual Environments (ICAT-EGVE). *Situational Awareness in Human Computer Interaction: Diana's World*.

2019 Outstanding reviewer, Conference on Empirical Methods in Natural Language Processing and International Joint Conference on Natural Language Processing (EMNLP-IJCNLP).

Invited Talks

2021 Krishnaswamy, N. (2021). Situated Grounding and Natural Language for Embodied AI. University of Colorado Boulder.

2018 Krishnaswamy, N. (2018). Grounded Linguistic Interaction in Multimodal Environments. Colorado State University.

2015 Krishnaswamy, N. (2015). Inside the Language Technology Revolution. San Juan College.

Prior Research Experience

2017-2020 *Postdoctoral Associate*, Brandeis University, Waltham, MA, USA
• DARPA-funded research and development in natural language understanding, artificial intelligence, and human-computer interaction.
• Project manager for lab work on DARPA Communicating With Computers (CwC) program, including grant and monthly report writing, running evaluations, packaging software and deliverables, and managing student research assistants and collaborators across multiple sites.

2013-2017 *Graduate Research Fellow*, Brandeis University, Waltham, MA, USA

Peer-Reviewed Publications

In the below list, * indicates authors who are/were students of mine, † indicates authors who are/were mentees of mine outside of a formal advising relationship, ‡ indicates where I am the primary author on a publication, and § indicates where I am the senior author on a publication. = indicates equally-contributing authors. Acceptance rates for conferences and workshops and impact factors for journals are given where available. These numbers may reflect the rate for the specific instance listed or an average over previous years. Acceptance rates for *strongly refereed* conferences (those with an acceptance rate <40%), have been bolded.

Book Chapters

2018 ‡Krishnaswamy, N. and Pustejovsky, J. (2018). Deictic Adaptation in a Virtual Environment. In *Spatial Cognition XI: International Conference on Spatial Cognition*. Springer. [Acceptance rate: 50%]

2016 ‡Krishnaswamy, N. and Pustejovsky, J. (2016). Multimodal Semantic Simulations of Linguistically Underspecified Motion Events. In *Spatial Cognition X: International Conference on Spatial Cognition*. Springer. [Acceptance rate: 55%]

Journal Articles

2023 †Henlein, A., *Gopinath, A., Krishnaswamy, N., Mehler, A., and Pustejovsky, J. (2023). Grounding Human-Object Interaction to Affordance Behavior in Multimodal Datasets. *Frontiers in Artificial Intelligence: Section Language and Computation*, 6. **[Impact factor: 3.0]** DOI: https://doi.org/10.3389/frai.2023.1084740

2022 ‡Krishnaswamy, N. and Pustejovsky, J. (2022). Affordance Embeddings for Situated Language Understanding. *Frontiers in Artificial Intelligence: Section Language and Computation*, 5. **[Impact factor: 3.0]** DOI: https://doi.org/10.3389/frai.2022.774752

2021 Pustejovsky, J. and Krishnaswamy, N. (2021). Embodied Human Computer Interaction. *KI - Künstliche Intelligenz: Special Issue on NLP and Semantics*, 35(3):307–327. **[Impact factor: 1.8]** DOI: https://doi.org/10.1007/s13218-021-00727-5

Pustejovsky, J. and Krishnaswamy, N. (2021). Situated Meaning in Multimodal Dialogue: Human-Robot and Human-Computer Interactions. *Traitement Automatique des Langues: Special Issue on Dialog and Dialog Systems*, 61(3):17–41.

Refereed Conference Proceedings

2023 *VanderHoeven, H., Blanchard, N., and §Krishnaswamy, N. (2023). Robust Motion Recognition using Gesture Phase Annotation. In *International Conference on Human-Computer Interaction (HCII)*. Springer. *(Accepted for publication.)* **[Avg. acceptance rate: 29%]**

†=Kandoi, C., †=Jung, C., *VanderHoeven, H., *Mannan, S., Krishnaswamy, N., and Blanchard, N. (2023). Intentional Microgesture Recognition for Extended Human-Computer Interaction. In *International Conference on Human-Computer Interaction (HCII)*. Springer. *(Accepted for publication.)* **[Avg. acceptance rate: 29%]**

=Krishnaswamy, N., =Oved, I., =Hartshorne, J. K., and =Pustejovsky, J. (2023). Meaning to Mean: A Precondition for Sentience and Understanding in Large Language Models. In *The Science of Consciousness (TSC)*. Center for Consciousness Studies. *(Accepted for presentation.)*

Weatherley, J., Dickler, R., Foltz, P. W., Srinivas, A., Pugh, S., Krishnaswamy, N., Whitehill, J., Bodzianowski, M., Perkoff, M., Southwell, R., Bush, J., Chang, M., Hirshfield, L. M., Showers, D., Ganesh, A., Li, Z., †Danilyuk, E., He, X., *Khebour, I., Dey, I., and D'Mello, S. K. (2023). The iSAT Collaboration Analytics Pipeline. In *International Learning Analytics and Knowledge Conference (LAK)*. Society for Learning Analytics Research. *(Accepted for presentation.)*

2022 *=Nath, A., †=Mahdipour Saravani, S., *Khebour, I., *Mannan, S., †Li, Z., and §Krishnaswamy, N. (2022). A Generalized Method for Automated Multilingual Loanword Detection. In *International Conference on Computational Linguistics (COLING)*. ACL. **[Acceptance rate: 33%]**

*Mannan, S. and §Krishnaswamy, N. (2022). Where am I and where should I go? Grounding positional and directional labels in a disoriented human balancing task. In *Conference on (Dis)embodiment*. ACL.

‡Krishnaswamy, N., †Pickard, W., *Cates, B., Blanchard, N., and Pustejovsky, J. (2022). The VoxWorld Platform for Multimodal Embodied Agents. In *International Conference on Language Resources and Evaluation (LREC)*. ACL. [Acceptance rate: 61%]

*Ghaffari, S. and §Krishnaswamy, N. (2022). Detecting and Accommodating Novel Types and Concepts in an Embodied Simulation Environment. In *Annual Conference on Advances in Cognitive Systems (ACS)*. Cognitive Systems Foundation.

‡Krishnaswamy, N. and *Ghaffari, S. (2022). Exploiting Embodied Simulation to Detect Novel Object Classes Through Interaction. In *Annual Meeting of the Cognitive Science Society (CogSci)*. Cognitive Science Society.

*Bradford, M., †Hansen, P., Beveridge, R., Krishnaswamy, N., and Blanchard, N. (2022). A deep dive into microphones for recording collaborative group work. In *International Conference on Educational Data Mining (EDM)*. International EDM Society.

Dickler, R., Foltz, P. W., Krishnaswamy, N., Whitehill, J., Weatherly, J., Bodzianowski, M., Perkoff, M., Southwell, R., Pugh, S., Bush, J., Chang, M., Hirshfield, L. M., Showers, D., Ganesh, A., Li, Z., †Danilyuk, E., He, X., *Khebour, I., Dey, I., Puntambekar, S., and D'Mello, S. K. (2022). iSAT speech-based AI display for small group collaboration in classrooms. In *Interactive event at International Conference on Artificial Intelligence in Education (AIEd)*. International AIEd Society.

Pustejovsky, J. and Krishnaswamy, N. (2022). Multimodal Semantics for Affordances and Actions. In *International Conference on Human-Computer Interaction (HCII)*. Springer. **[Avg. acceptance rate: 29%]**

2021 Pustejovsky, J. and Krishnaswamy, N. (2021). The Role of Embodiment and Simulation in Evaluating HCI: Theory and Framework. In *International Conference on Human-Computer Interaction (HCII)*. Springer. **[Avg. acceptance rate: 29%]**

‡Krishnaswamy, N. and Pustejovsky, J. (2021). The Role of Embodiment and Simulation in Evaluating HCI: Experiments and Evaluation. In *International Conference on Human-Computer Interaction (HCII)*. Springer. **[Avg. acceptance rate: 29%]**

2020 ‡Krishnaswamy, N. and Pustejovsky, J. (2020). Neurosymbolic AI for Situated Language Understanding. In *Annual Conference on Advances in Cognitive Systems (ACS)*. Cognitive Systems Foundation. [Acceptance rate: 41%]

‡Krishnaswamy, N. and Pustejovsky, J. (2020). A Formal Analysis of Multimodal Referring Expressions Under Common Ground. In *International Conference on Language Resources and Evaluation (LREC)*. ACL. [Acceptance rate: 60%]

‡Krishnaswamy, N., Narayana, P., Bangar, R., Rim, K., Patil, D., McNeely-White, D. G., Ruiz, J., Draper, B., Beveridge, R., and Pustejovsky, J. (2020). Diana's World: A Situated Multimodal Interactive Agent. In *AAAI Conference on Artificial Intelligence (AAAI): Demos Program*. AAAI.

‡Krishnaswamy, N., Beveridge, R., Pustejovsky, J., Patil, D., McNeely-White, D. G., Wang, H., and Ortega, F. R. (2020). Situational Awareness in Human Computer Interaction: Diana's World. In *International Conference on Artificial Reality and Telexistence & Eurographics Symposium on Virtual Environments (ICAT-EGVE): Demos*. ACM/Eurographics.

Pustejovsky, J. and Krishnaswamy, N. (2020). Embodied Human-Computer Interactions through Situated Grounding. In *International Conference on Intelligent Virtual Agents (IVA)*. ACM. **[Acceptance rate: 26%]**

†Hutchens, M., Krishnaswamy, N., Cochran, B., and Pustejovsky, J. (2020). Jarvis: A Multimodal Visualization Tool for Bioinformatic Data. In *International Conference on Human-Computer Interaction (HCII)*. Springer. **[Avg. acceptance rate: 29%]**

†Krajovic, K., Krishnaswamy, N., Dimick, N. J., Salas, R. P., and Pustejovsky, J. (2020). Situated Multimodal Control of a Mobile Robot: Navigation through a Virtual Environment. In *Special Session on Situated Dialogue with Virtual Agents and Robots (RoboDIAL): Late-Breaking Papers*. Non-archival.

2019 ‡Krishnaswamy, N. and Pustejovsky, J. (2019). Generating a Novel Dataset of Multimodal Referring Expressions. In *International Workshop on Computational Semantics (IWCS)*. ACL. [Acceptance rate: 43%]

‡Krishnaswamy, N., Friedman, S., and Pustejovsky, J. (2019). Combining Deep Learning and Qualitative Spatial Reasoning to Learn Complex Structures from Sparse Examples with Noise. In *AAAI Conference on Artificial Intelligence (AAAI)*. AAAI. **[Acceptance rate: 16%]**

McNeely-White, D., Ortega, F., Beveridge, R., Draper, B., Bangar, R., Patil, D., , Pustejovsky, J., Krishnaswamy, N., Rim, K., Ruiz, J., and Wang, I. (2019). User-Aware Shared Perception for Embodied Agents. In *International Conference on Humanized Computing and Communication (HCC)*. IEEE.

2018 ‡Krishnaswamy, N. and Pustejovsky, J. (2018). An Evaluation Framework for Multimodal Interaction. In *International Conference on Language Resources and Evaluation (LREC)*. ACL. [Acceptance rate: 65%]

Narayana, P., Krishnaswamy, N., Wang, I., Bangar, R., Patil, D., Mulay, G., Rim, K., Beveridge, R., Ruiz, J., Pustejovsky, J., and Draper, B. (2018). Cooperating with Avatars Through Gesture, Language and Action. In *Intelligent Systems Conference (IntelliSys)*. IEEE. **[Acceptance rate: 34%]**

Do, T., Krishnaswamy, N., Rim, K., and Pustejovsky, J. (2018). Multimodal Interactive Learning of Primitive Actions. In *AAAI Fall Symposium: Artificial Intelligence for Human-Robot Interaction*. AAAI.

Do, T., Krishnaswamy, N., and Pustejovsky, J. (2018). Teaching Virtual Agents to Perform Complex Spatial-Temporal Activities. In *AAAI Spring Symposium: Integrating Representation, Reasoning, Learning, and Execution for Goal Directed Autonomy*. AAAI.

2017 [‡]Krishnaswamy, N., Narayana, P., Wang, I., Rim, K., Bangar, R., Patil, D., Mulay, G., Ruiz, J., Beveridge, R., Draper, B., and Pustejovsky, J. (2017). Communicating and Acting: Understanding Gesture in Simulation Semantics. In *International Workshop on Computational Semantics (IWCS)*. ACL. [Acceptance rate: 53%]

Pustejovsky, J., Krishnaswamy, N., and Do, T. (2017). Object Embodiment in a Multimodal Simulation. In *AAAI Spring Symposium: Interactive Multisensory Object Perception for Embodied Agents*. AAAI.

2016 [‡]Krishnaswamy, N. and Pustejovsky, J. (2016). VoxSim: A Visual Platform for Modeling Motion Language. In *International Conference on Computational Linguistics (COLING): Technical Papers*. ACL. **[Acceptance rate: 32%]**

Pustejovsky, J. and Krishnaswamy, N. (2016). Visualizing Events: Simulating Meaning in Language. In *Annual Meeting of the Cognitive Science Society (CogSci)*. Cognitive Science Society.

Pustejovsky, J. and Krishnaswamy, N. (2016). VoxML: A Visualization Modeling Language. In *International Conference on Language Resources and Evaluation (LREC)*. ACL. [Acceptance rate: 60%]

Refereed Workshop Proceedings

2023 Dey, I., Puntambekar, S., Li, R., Gengler, D., Dickler, R., Hirshfield, L. M., Clevenger, C., Rose, S., [*]Bradford, M., and Krishnaswamy, N. (2023). The NICE framework: Analyzing Students' Nonverbal Interactions During Collaborative Learning. In *Interactive Workshop: Collaboration Analytics*. Society for Learning Analytics Research. *(Accepted for publication.)*

2022 [*]Nath, A., [†]Ghosh, R., and [§]Krishnaswamy, N. (2022). Phonetic, Semantic, and Articulatory Features in Assamese-Bengali Cognate Detection. In *Workshop on NLP for Similar Languages, Varieties, and Dialects (VarDial)*. ACL. [Acceptance rate: 54%]

[*]Tomar, A. and [§]Krishnaswamy, N. (2022). Exploring Correspondences Between Gibsonian and Telic Affordances for Object Grasping. In *Workshop on Annotation, Recognition and Evaluation of Actions (AREA)*. ACL.

[*]Bradford, M., [†]Hansen, P., Lai, K., Brutti, R., Dickler, R., Hirshfield, L. M., Pustejovsky, J., Blanchard, N., and [§]Krishnaswamy, N. (2022). Challenges and Opportunities in Annotating a Multimodal Collaborative Problem Solving Task. In *Interdisciplinary Approaches to Getting AI Experts and Education Stakeholders Talking Workshop (Bridging AIEd)*. International AIEd Society.

[†]Castillon, I., [†]Venkatesha, V., [*]VanderHoeven, H., [*]Bradford, M., Krishnaswamy, N., and Blanchard, N. (2022). Multimodal Features for Group Dynamic-Aware Agents. In

Interdisciplinary Approaches to Getting AI Experts and Education Stakeholders Talking Workshop (Bridging AIEd). International AIEd Society.

2021 ‡Krishnaswamy, N. and *Alalyani, N. (2021). Embodied Multimodal Agents to Bridge the Understanding Gap. In *Workshop on Bridging Human-Computer Interaction and Natural Language Processing (HCI+NLP)*. ACL.

2020 Pustejovsky, J., Krishnaswamy, N., Beveridge, R., Ortega, F. R., Patil, D., Wang, H., and McNeely-White, D. G. (2020). Interpreting and Generating Gestures with Embodied Human-Computer Interactions. In *Workshop on Generation and Evaluation of Non-Verbal Behaviour for Embodied Agents (GENEA)*. ACM.

2019 ‡Krishnaswamy, N. and Pustejovsky, J. (2019). Situated Grounding Facilitates Multimodal Concept Learning for AI. In *Visually Grounded Interaction and Language Workshop (ViGIL)*. Neural Information Processing Systems Foundation.

‡Krishnaswamy, N. and Pustejovsky, J. (2019). Multimodal Continuation-style Architectures for Human-Robot Interaction. In *Workshop on Cognitive Vision: Integrated Vision and AI for Embodied Perception and Interaction*. Cognitive Systems Foundation.

Pustejovsky, J. and Krishnaswamy, N. (2019). Situational Grounding within Multimodal Simulations. In *AAAI Workshop on Games and Simulations in AI (GameSim)*. AAAI.

2018 ‡Krishnaswamy, N., Do, T., and Pustejovsky, J. (2018). Learning Actions from Events Using Agent Motions. In *Workshop on Annotation, Recognition and Evaluation of Actions (AREA)*. ACL.

Pustejovsky, J. and Krishnaswamy, N. (2018). The Role of Event Simulation in Spatial Cognition. In *Workshop on Models and Representations in Spatial Cognition (MRSC)*. Springer.

Pustejovsky, J. and Krishnaswamy, N. (2018). Every Object Tells a Story. In *Workshop on Events and Stories in the News (EventStory)*. ACL.

2017 ‡Krishnaswamy, N. and Pustejovsky, J. (2017). Do You See What I See? Effects of POV on Spatial Relation Specifications. In *International Workshop on Qualitative Reasoning (QR)*. AAAI/International Joint Conferences on Artificial Intelligence.

Pustejovsky, J., Krishnaswamy, N., Draper, B., Narayana, P., and Bangar, R. (2017). Creating Common Ground Through Multimodal Simulations. In *Workshop on Foundations of Situated and Multimodal Communication (FSMC)*. ACL.

2016 Pustejovsky, J., Krishnaswamy, N., Do, T., and Kehat, G. (2016). The Development of Multimodal Lexical Resources. In *Workshop on Grammar and the Lexicon (GramLex)*. ACL.

Do, T., Krishnaswamy, N., and Pustejovsky, J. (2016). ECAT: Event Capture Annotation Tool. In *International Workshop on Semantic Annotation (ISA)*. ACL.

2014 Pustejovsky, J. and Krishnaswamy, N. (2014). Generating Simulations of Motion Events from Verbal Descriptions. In *Lexical and Computational Semantics (*SEM)*. ACL.

2 **Teaching**

Teaching Experience

Term	Course	Enrollment	Audience
Fall 2022	CS542: Natural Language Processing	34 (17 in-person, 17 online)	Graduate
Spring 2022	CS445: Introduction to Machine Learning	86 (75 in-person, 11 online)	Undergraduate
Fall 2021	CS542: Natural Language Processing	30 (22 in-person, 8 online)	Graduate
Spring 2021	CS445: Introduction to Machine Learning	67 (55 in-person, 12 online)	Undergraduate

Other Teaching Activities

2022 Instructor, Grounding Meaning Representation for Situated Reasoning. Tutorial, Meeting of the Asia-Pacific Chapter of the Association for Computational Linguistics and International Joint Conference on Natural Language Processing (AACL-IJCNLP). Taipei, ROC. November, 2022.

Instructor, Multimodal Semantics for Affordances and Actions. European Summer School in Logic, Language, and Information (ESSLLI). Galway, Ireland. August, 2022.

2018-2020 Staff tutor, Computational Linguistics Master's program. Brandeis University.
• Tutored MS students in depth on coursework in multiple courses, including COSI 114: Fundamentals of Computational Linguistics, and COSI 134: Statistical Approaches to Natural Language Processing.

2017 Instructor, Building Multimodal Simulations for Natural Language. Tutorial, Meeting of the European Chapter of the Association for Computational Linguistics (EACL). Valencia, Spain. April, 2017.

3 **Advising/Supervision**

Graduated Ph.D. Students

2020 Co-advisor. Patil, D. K. (2022). *Something is fishy!—How ambiguous language affects generalization of video action recognition networks*. PhD thesis, Colorado State University.

Graduated M.S.. Students

2022 Advisor. Nath, A. (2022). *Linear Mappings: Semantic Transfer from Transformer Models for Cognate Detection and Coreference Resolution*. Master's thesis, Colorado State University.

Advisor. Garcia, J. S. (2022). *Applications of Topological Data Analysis to Natural Language Processing and Computer Vision*. Master's thesis, Colorado State University.

Advisor. Patchava, R. S. (2022). *Evaluating Interchangeability of Face Feature Vectors*. Master's project, Colorado State University.

Advisor. Pandya, K. A. (2022). *Qualitative Spatial Relation Representation with ML Embedding Spaces*. Master's project, Colorado State University.

2021 Advisor. Mogullapalli, S. (2021). *Mapping Between Face Recogniton Feature Vector from Various CNN Models*. Master's project, Colorado State University.

Advisor. Gaddam, S. (2021). *Exploring Embedding Spaces in Transformers by Mapping Feature Vectors*. Master's project, Colorado State University.

Advisor. Katragadda, M. (2021). *Distributed Training of 3D Object Recognition Using Point Clouds*. Master's project, Colorado State University.

2018 Reader. Storozum, J. (2018). *Opposites Attract—Or Do They?: Investigating Negated Verbs in Distributional Semantic Space*. Master's thesis, Brandeis University.

Current Ph.D. Students

Nada Alalyani
Mariah Bradford (co-advised with N. Blanchard)
Brittany Cates
Anju Gopinath
Sadaf Ghaffari
Ibrahim Khebour
Shadi Manafi
Sheikh Mannan
Abhijnan Nath

Current M.S. Students

Aniket Tomar
Hannah VanderHoeven (co-advised with N. Blanchard)

Current Undergraduate Mentees

Quincy Meisman

4 Academic Service

2023 Program committee, Meeting of the Association for Computational Linguistics (ACL). Toronto, ON, Canada.

Program committee, Meeting of the Cognitive Science Society (CogSci). Sydney, NSW, Australia.

Program committee, AAAI Conference on Artificial Intelligence (AAAI). Washington, DC, USA.

Program committee, Conference on Learning from Small Data (LSD). Gothenburg, Sweden.

On-call reviewer for ACL Rolling Review (ACL, NAACL, EMNLP).

2022 Area chair, Multimodal NLP, Grounded Language Acquisition, and Human–Robot Interaction, International Conference on Computational Linguistics (COLING). Gyeongju, Korea.

Ethics board, International Conference on Computational Linguistics (COLING). Gyeongju, Korea.

Program committee, Dialogue and Interactive Systems and Speech and Multimodality Processing tracks, Meeting of the Asian Chapter of the Association for Computational Linguistics and International Joint Conference on Natural Language Processing (AACL-IJCNLP). Taipei, ROC.

Program committee, AAAI Conference on Artificial Intelligence (AAAI). Virtual meeting (Hosted: Vancouver, BC, Canada).

Program committee, Meeting of the Cognitive Science Society (CogSci). Toronto, ON, Canada.

Program committee, Conference on (Dis)embodiment. Gothenburg, Sweden.

Program committee, Workshop on Annotation, Recognition and Evaluation of Actions (AREA). Galway, Ireland.

On-call reviewer for ACL Rolling Review (ACL, NAACL, EMNLP).

Journal reviews: *Frontiers in Neurorobotics*, *Human-Computer Interaction*, *Artificial Intelligence Review*, *International Journal of Human-Computer Interaction*.

2021 Organizing committee, Beyond Language: Workshop on Multimodal Semantic Representation (MMSR). Virtual meeting (Hosted: Groningen, The Netherlands).

Program committee, Goal Reasoning Workshop (GRW). Virtual meeting (Hosted: Dayton, OH, USA).

Program committee, Conference on Reasoning in Interaction (ReInAct). Gothenburg, Sweden.

Program committee, Language Grounding to Vision, Robotics and Beyond track, Meeting of the Association for Computational Linguistics and International Joint Conference on Natural Language Processing (ACL-IJCNLP). Virtual meeting (Hosted: Bangkok, Thailand).

Program committee, International Workshop on Computational Semantics (IWCS). Virtual meeting (Hosted: Groningen, The Netherlands).

Program committee, Meeting of the Cognitive Science Society (CogSci). Vienna, Austria.

Program committee, Language Grounding to Vision, Robotics and Beyond track, Meeting of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT). Virtual meeting (Hosted: Mexico City, Mexico).

Program committee, Language Grounding to Vision, Robotics and Beyond track, Meeting of the European Chapter of the Association for Computational Linguistics (EACL).

Virtual meeting (Hosted: Kyiv, Ukraine).

Program committee, AAAI Conference on Artificial Intelligence (AAAI). Virtual meeting (Hosted: San Francisco, CA, USA).

Program committee, Joint ACL-ISO Workshop on Interoperable Semantic Annotation (ISA). Virtual meeting (Hosted: Groeningen, The Netherlands).

2020 Program committee, Meeting of the Asia-Pacific Chapter of the Association for Computational Linguistics and International Joint Conference on Natural Language Processing (AACL-ICJNLP). Virtual meeting (Hosted: Suzhou, China).

Program committee, Joint ACL-ISO Workshop on Interoperable Semantic Annotation (ISA).

Program committee, Language Grounding to Vision, Robotics and Beyond area, Conference on Empirical Methods in Natural Language Processing (EMNLP). Virtual meeting (Hosted: Punta Cana, Dominican Republic).

Program committee, Meeting of the Cognitive Science Society (CogSci). Virtual meeting (Hosted: Toronto, ON, Canada).

Program committee, Language Grounding to Vision, Robotics and Beyond area, Meeting of the Association for Computational Linguistics (ACL). Virtual meeting (Hosted: Seattle, WA, USA).

Program committee, AAAI Conference on Artificial Intelligence (AAAI). New York, NY, USA.

Program committee, Student session, Web Summer School on Logic, Language, and Information (WeSSLLI). Virtual meeting (Hosted: Waltham, MA, USA).

Program committee, Special session on Gestures and Natural Language Semantics: Investigations at the Interface at *Sinn und Bedeutung* (SuB25-Gestures). London, England, UK.

Journal Reviews: IEEE *Transactions on Cognitive and Developmental Systems*.

2019 Program committee, Speech, Vision, Robotics, Multimodal and Grounding area (long and short papers), Conference on Empirical Methods in Natural Language Processing and International Joint Conference on Natural Language Processing (EMNLP-IJCNLP). Hong Kong.

Program committee, Vision, Robotics, Multimodal, Grounding and Speech area, Meeting of the Association for Computational Linguistics (ACL). Florence, Italy.

Program committee, Meeting of the Cognitive Science Society (CogSci). Montréal, QC, Canada.

Program committee, Semantics area (long papers), Meeting of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT). Minneapolis, MN, USA.

2018 Program committee, Workshop on Dialogue and Perception (DaP). Gothenburg, Sweden.

Pre-submission mentor, Student Research Workshop, Meeting of the North American Chapter of the Association for Computational Linguistics (NAACL). New Orleans, LA, USA.

Program committee, Workshop on Annotation, Recognition and Evaluation of Actions (AREA). Miyazaki, Japan.

Professional Membership

Association for Computational Linguistics (ACL), Association for the Advancement of Artificial Intelligence (AAAI), Linguistic Society of America (LSA), European Language Resource Association (ELRA).